

Introduction to Microsoft Excel for ZOS

1 What is MS Excel?

MS Excel is a program you can use to create and format spreadsheets and analyze information. In this course, we use Excel to model and run simulations.

2 What does it look like?

An Excel-file contains sheets built up of cells:

	A	B	C	D	E	F	G	H
1	A1	B1	etc.					
2	A2	B2	etc.					
3	etc.	etc.	etc.					
4								H4

Columns: A, B, ..., Z, AA, ..., AZ, BA, ..., BZ, CA, ..., IA, ..., IV

Rows: 1, 2, ..., 65536

3 How to perform calculations?

To perform calculations with numbers in the cells of an Excel sheet, you need to enter a formula in the cell you want the result in. Every formula starts with the = sign. The table below shows common functions that are used in the exercises of part A of the reader. There are many more functions that can be found in the menu Insert → Function.

Function	Description	Example
cell number	The value of a another cell	=A2
+	Addition	=A2+B2
−	Subtraction	=A2−B2
*	Multiplication	=A2*B2
/	Division	=A2/B2
^	Exponentiation	=A2^B2
()	Brackets	=A2+B2*C2 or =(A2+B2)*C2
SUM(number1, number2, ...)	Sum of all <i>number</i> values. The values can be numbers and formulas. A colon means 'to', see the second example: from A2 to A5.	=SUM(A2, A3, A4, A5) =SUM(A2:A5)

IF(logical-test; value-if-true; value-if-false)	A logical if-then-else rule: • If <i>logical-test</i> is true, then the cell is filled with <i>value-if-true</i>. • If <i>logical-test</i> is false, then the cell is filled with <i>value-if-false</i>. The values for <i>value-if-true</i> and <i>value-if-false</i> can be text (with quotes around it), numbers or formulas. The value for <i>logical-test</i> can be a number or a formula.	=IF(A2 = 2; B2; C2) =IF(AND(A2; B2); 1; 0) =IF(A2 > 3; "yes"; "no")
AND(logical1; logical2;...)	If all logical statements are true, the result is TRUE. Otherwise, the result is FALSE. This function can consist of one or more logical statements (= formulas).	AND(A2; B2)
OR(logical1; logical2; ...)	If at least one of the logical statements is true, the result is TRUE. If none are true, the result is FALSE.	OR(A2 = 1; B2 = 1; C2 = 1)
NOT(logical)	The result is TRUE if the logical statement is false, the result is FALSE if the logical statement is true.	NOT(A2)

4 Tips & tricks

Below are a few tips and tricks that make life easier when using Excel.

4.1 Filling cells

When you want to repeat the same number, an increasing or decreasing number or a formula in a number of cells in the same row or column, you do not need to copy and paste every cell or retype the number or formula. There are three different ways to automatically fill a row or column. Use the examples below to select the method that fits your action best.

1. Select the cell you want to copy. Drag the little square in the right lower corner to select all cells you want to fill.
2. Click the cell you want to copy, hold down *shift* and click the last of all the cells you want to fill. Note that all cells need to be adjacent. Go to Edit → Fill → Down / Right / Up / Left. You can also use the short keys for the actions down and right: *ctrl+d* and *ctrl+r*.
3. Select the cell you want to copy (and all the cells you want to fill, see method 2). Go to Edit → Fill → Series and choose how you want to fill the selection. Use a stop value if you have not selected all cells.

	Result if original cell contains a	
Method	formula with cell name(s)	number
1 and 2	The formula is copied to all cells. If the formula contains another cell name, that cell name is automatically updated (see example below). If you do not want the cell name to be updated, use the \$ sign before the letter, number, or both (\$A2, A\$2 or \$A\$2).	The same number is entered in all cells.
3	The result of the formula is treated as a number.	Depends on the selected options. Most often used for increasing numbers.

Example 1, method 1: I have entered the following formula in cell C4: =B3. I want the same formula in the cells below C4 and every cell needs to refer back to the cell one up and one left. Therefore, I use method 1 and drag the little black square to the last cell of my column. The column is filled with the following formulas:

	C
4	=B3
5	=B4
6	=B5
7	=B6

Example 2, method 2: I have entered the number 400 in cell B2. Now I want to fill the next 9 cells of that column with the same number. I select cell B2, hold down *shift* and select cell B11, cells B2 to B11 are now selected. I go to Edit → Fill → Down (or I click *ctrl+d*). The selected cells all have the number 400.

Example 3, method 3: I want to enter the numbers 0 to 100 in a column. I enter 0 in the first cell, select that cell and go to Edit → Fill → Series. I select Columns, Linear, Step value 1 and Stop value 100.

4.2 Inserting and deleting rows or columns

Right-click the letter above a column or the number left to a row and select Insert or Delete. The letters and numbers are automatically updated.

4.3 Inserting a chart

1. Select the data you want in the graph, possibly including the name of a column and/or the scale of your data (numbers on the x-axis).
2. Go to Insert → Chart
3. The charts used in this course are all line charts, so choose Line. The top left chart is a line without markers.
4. Click Next.
5. If you selected a column name, it is automatically detected as name. If not, you can enter a name on the tab Series.
6. Click Next.

7. Here you can enter names for the chart and the axes and many more options. After setting them, click Finish.

4.4 Locking headings

If the columns of your Excel sheet are very long, the headers of the columns disappear if you scroll down. To freeze the headings, select the row below the header and go to Window → Freeze panes. Your header is now locked.

5 Questions?

You can use the Help function in Excel to find an answer to your question. During the 'werkcollege' teachers and/or teaching assistants are available to answer any questions you may have about Excel or about the exercises. Or you can always send an email to one of the teachers of the course.